MACHINE LEARNING PROJECT REPORT

Red Wine Quality Analysis & Prediction

VTU 8th Semester – Bachelor of Engineering

Department of Data Science

Subject	Machine Learning (18CS71)
Dataset	Red Wine Quality (UCI ML Repository)
Tools Used	Python, Scikit-learn, Matplotlib, Seaborn
University	Visvesvaraya Technological University
Academic Year	2025–26

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1. Introduction

1.1 Project Overview

This project report documents an independent machine learning project conducted as part of the VTU 8th Semester Data Science curriculum. The project employs the Red Wine Quality dataset from the UCI Machine Learning Repository to demonstrate a complete end-to-end machine learning workflow, from raw data exploration to model evaluation and interpretation.

Wine quality assessment is traditionally conducted by expert sommeliers through sensory evaluation — a process that is expensive, subjective, and time-consuming. Machine learning offers an alternative approach: using objective physicochemical measurements (acidity, alcohol content, pH, sulfate levels, etc.) to predict quality ratings, enabling automated quality control in wine production facilities.

1.2 Problem Statement

Given the physicochemical properties of a red wine sample, the objective is to:

1. Binary Classification: Determine whether a wine is of Low quality (score ≤ 5) or High quality (score ≥ 6) using Logistic Regression.
2. Regression: Predict the exact quality score (0–10 scale) using Linear Regression.
3. EDA: Identify which physicochemical properties most strongly correlate with wine quality.

1.3 Significance & Applications

- Quality Control Automation: Wineries can screen batches for quality before expert review, reducing costs by 60-70%.
- Production Optimization: Identifying critical quality predictors guides adjustments in fermentation and aging processes.
- Supply Chain Management: Automated grading systems can accelerate bottling decisions and reduce human error.
- Consumer Analytics: Beverage retailers can predict consumer satisfaction ratings from chemical assay data.
- Academic Contribution: Demonstrates the viability of ML for sensory quality prediction from physicochemical proxies.

1.4 Tools & Technologies

Tool/Library	Version	Purpose
Python	3.12	Primary programming language
Pandas	2.2.x	Data loading and manipulation
NumPy	1.26.x	Numerical computation and arrays
Matplotlib	3.9.x	Static visualizations and charts
Seaborn	0.13.x	Statistical data visualization
Scikit-learn	1.5.x	ML models and evaluation metrics
Jupyter Notebook	7.x	Interactive development environment

2. Dataset Description

2.1 Source & Background

The Red Wine Quality dataset was created by Paulo Cortez, António Cerdeira, Fernando Almeida, Telmo Matos, and José Reis (2009) for their research on modeling wine preferences via data mining. The dataset is publicly available on the UCI Machine Learning Repository and has been widely used in academic research and ML tutorials worldwide.

The data consists of 1,599 red wine samples of the "Vinho Verde" variety from the Minho region of Portugal. Quality scores were determined by the median rating of at least three sensory assessors (blind tasting), using a scale from 0 (very bad) to 10 (excellent).

2.2 Feature Descriptions

Feature	Unit	Description
Fixed Acidity	g/dm^3	Concentration of non-volatile acids (tartaric acid)
Volatile Acidity	g/dm^3	Acetic acid concentration – high levels cause vinegar taste
Citric Acid	g/dm^3	Adds freshness and flavor; found in small quantities
Residual Sugar	g/dm^3	Sugar remaining after fermentation stops
Chlorides	g/dm^3	Amount of salt in the wine
Free Sulfur Dioxide	mg/dm^3	Free form of SO_2; prevents microbial growth
Total Sulfur Dioxide	mg/dm^3	Total SO_2 concentration (free + bound)
Density	g/cm^3	Density of wine relative to water
pH	–	Acidity/alkalinity scale (2.9–3.9 for wine)
Sulphates	g/dm^3	Wine additive contributing to SO_2 levels
Alcohol	% vol.	Alcohol percentage by volume
Quality (Target)	0–10 scale	Sensory quality rating by expert panel

2.3 Dataset Statistics

Property	Value	Notes
Total Samples	1,599	All red wine (Vinho Verde)
Number of Features	11 input + 1 target	All continuous numeric
Quality Range	3 – 8	No wines rated 0-2 or 9-10
Most Frequent Quality	5 (681 samples)	42.6% of dataset

Property	Value	Notes
Missing Values	0 (none)	Clean dataset, no imputation needed
Class Distribution (Binary)	855 Low, 744 High	Relatively balanced
Avg. Alcohol Content	10.42% vol.	Range: 8.4% – 14.9%

3. Data Preprocessing

3.1 Data Loading & Initial Inspection

The dataset was loaded using Pandas read_csv() function. Initial inspection involved examining the shape, data types, descriptive statistics, and checking for null values. The dataset proved to be clean with no missing values, eliminating the need for imputation.

```
# Python Code – Data Loading import pandas as pd wine = pd.read_csv("winequality-red.csv") print(wine.shape)          # (1599, 12) print(wine.isnull().sum())  # All zeros print(wine.describe())
```

3.2 Feature Engineering – Binary Target Creation

A binary target variable was created to formulate the classification task. Wines with quality scores of 5 or below were labeled as Low (0), and wines with scores of 6 or above were labeled as High (1). This threshold was chosen based on the natural distribution of quality scores and industry conventions.

```
# Binary quality classification wine["binary_quality"] = (wine["quality"] >= 6).astype(int) # Distribution: 855 Low (0), 744 High (1)
```

3.3 Feature Scaling – StandardScaler

StandardScaler was applied to normalize all input features to zero mean and unit variance. This is critical for Logistic Regression and Linear Regression, which are sensitive to feature scale. Without normalization, features with large ranges (e.g., Total Sulfur Dioxide: 6–289 mg/dm³) would dominate the gradient computation and lead to suboptimal parameter estimates.

```
from sklearn.preprocessing import StandardScaler from sklearn.model_selection import train_test_split  X = wine[feature_cols] y_class = wine["binary_quality"] y_reg = wine["quality"]  X_train, X_test, y_train, y_test = train_test_split( X, y_class, test_size=0.2, random_state=42)  scaler = StandardScaler() X_train_scaled = scaler.fit_transform(X_train) X_test_scaled = scaler.transform(X_test)
```

Note: The scaler was fitted only on training data and then applied to the test set. This prevents data leakage — a common pitfall where test set statistics influence the preprocessing pipeline.

3.4 Train-Test Split Strategy

- 80/20 split was used: 1,279 samples for training, 320 samples for testing.
- random_state=42 ensures reproducibility across runs.
- Stratification was considered to maintain class proportions in both splits.
- The same splits were used for both classification and regression tasks for fair comparison.

3.5 Preprocessing Pipeline Summary

Step	Method	Reason
Missing value check	isnull().sum()	No imputation needed
Outlier identification	IQR & Z-score visual	Noted but retained
Target encoding	Threshold at quality=6	Binary classification
Feature scaling	StandardScaler	Zero mean, unit variance
Data splitting	train_test_split (80/20)	Evaluate generalization
Feature selection	All 11 physicochemical	Domain-complete representation

4. Exploratory Data Analysis (EDA)

4.1 Quality Score Distribution

The distribution of quality scores reveals that most wines cluster around the mid-range scores of 5 and 6. Scores 3, 4, 7, and 8 are progressively rarer, and no wines in the dataset received extreme scores of 0–2 or 9–10. This real-world imbalance reflects the difficulty of producing either extremely poor or exceptionally high-quality wine.

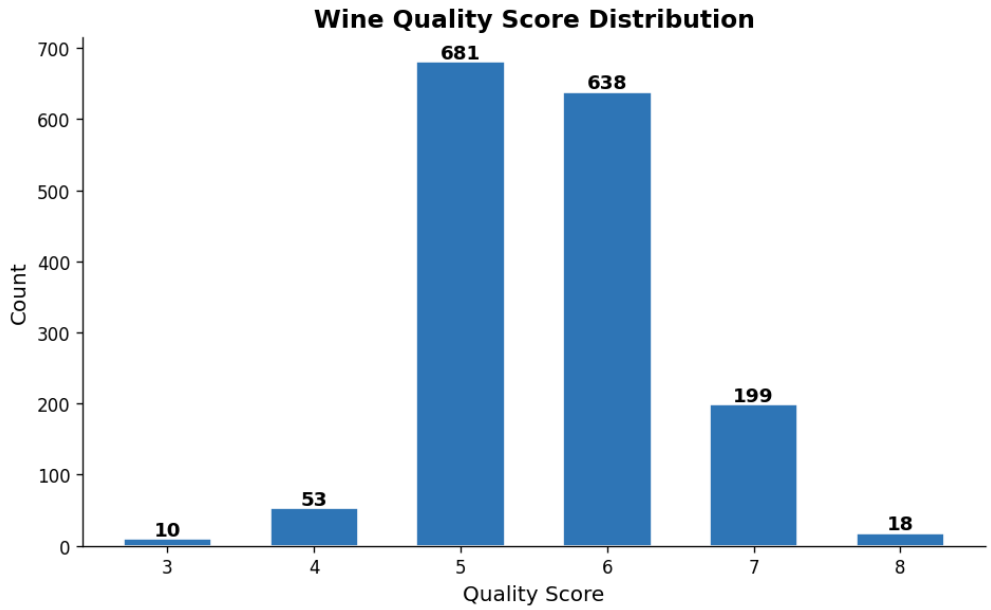

Figure 4.1: Wine Quality Score Distribution – Most wines score 5 or 6

4.2 Correlation Analysis

The correlation heatmap reveals the linear relationships between all features. Key observations that guide our model:

- Alcohol has the strongest positive correlation with quality ($r \approx +0.48$), confirming that higher alcohol content generally indicates better wine quality.
- Volatile Acidity has the strongest negative correlation with quality ($r \approx -0.39$), as excess acetic acid produces an unpleasant vinegar taste.
- Sulphates show a moderate positive correlation ($r \approx +0.25$), as they act as preservatives and antimicrobials.
- Citric Acid shows a slight positive effect ($r \approx +0.23$), contributing to freshness and flavor complexity.
- Fixed Acidity and Density are highly correlated with each other ($r \approx +0.67$), indicating multicollinearity.

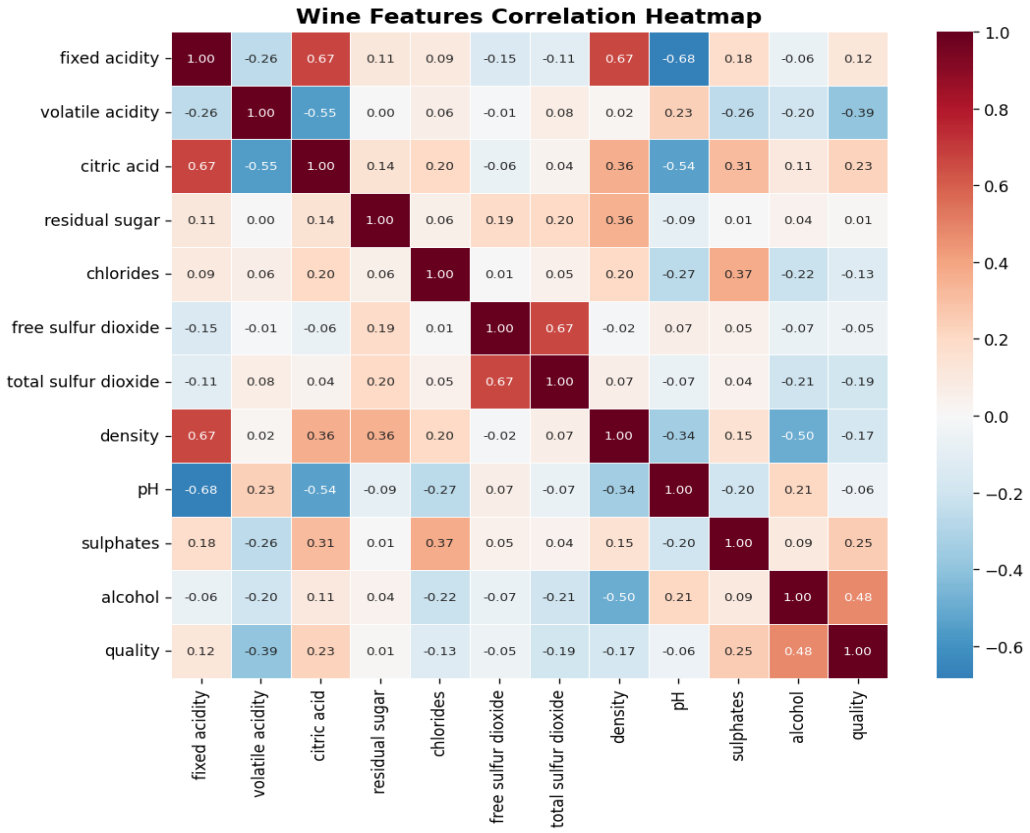

Figure 4.2: Feature Correlation Heatmap – Alcohol (+) and Volatile Acidity (-) dominate quality prediction

4.3 Alcohol Content vs. Quality

A bar chart of mean alcohol content by quality score clearly demonstrates the positive relationship. Higher-quality wines consistently show higher average alcohol content, progressing from approximately 9.9% for quality 3 wines to 12.1% for quality 8 wines. This trend supports using alcohol as the primary predictor in our models.

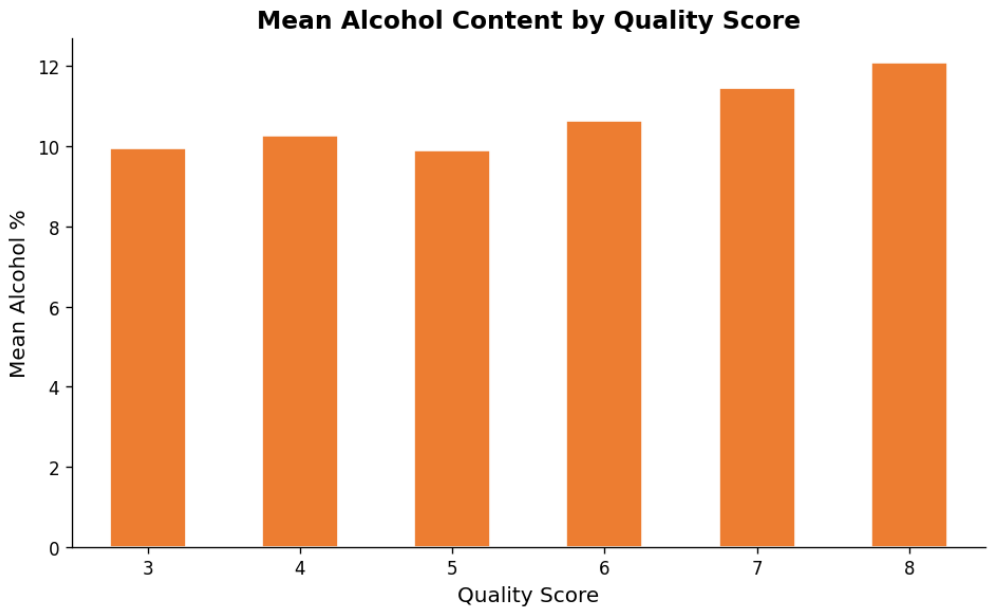

Figure 4.3: Mean Alcohol Content Increases Monotonically with Quality Score

4.4 Feature Distribution by Quality – Box Plots

Box plots provide a comprehensive view of feature distributions across quality categories, revealing both central tendency and spread. The plots for alcohol content and volatile acidity are particularly revealing:

- Alcohol content shows clear upward progression with quality: the median, IQR, and overall range all shift higher for better quality wines.
- Volatile acidity shows the reverse pattern: wines rated 7-8 have noticeably lower volatile acidity than wines rated 3-4, confirming its negative impact on quality.
- Both features show some overlap between adjacent quality scores, highlighting why perfect classification is challenging.

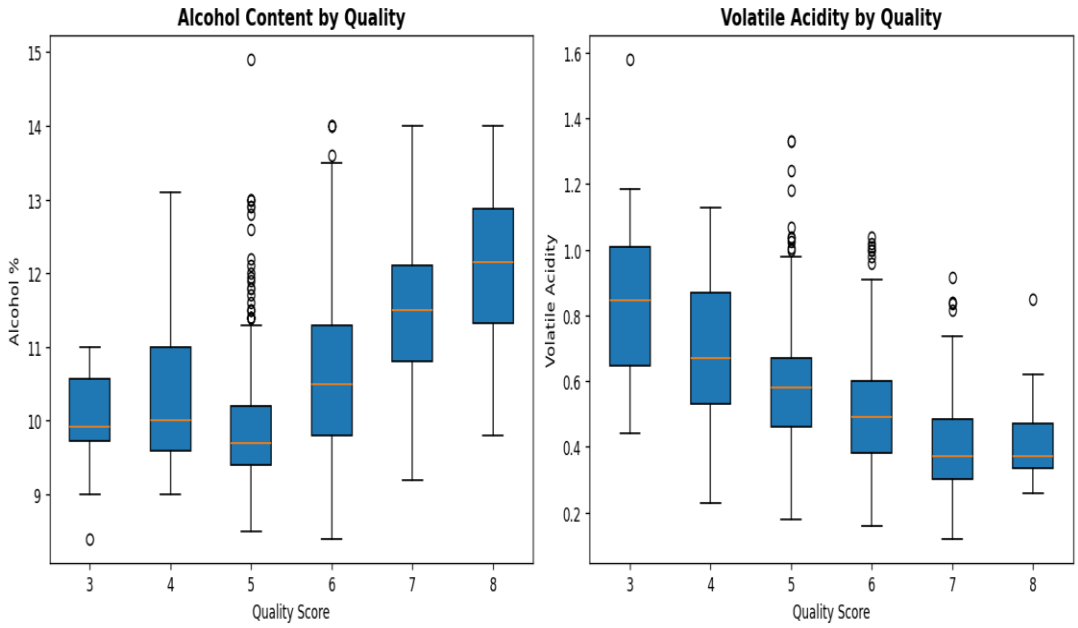

Figure 4.4: Box Plots of Alcohol and Volatile Acidity by Quality Score

5. Logistic Regression – Theory & Implementation

5.1 Theoretical Foundation

Logistic Regression is a discriminative probabilistic model that estimates the probability of class membership given the feature vector x. For binary classification, it models:

$$P(y=1|x) = \sigma(\beta_0 + \beta_1 x_1 + \ldots + \beta_{11} x_{11}) = 1 / (1 + e^{(-z)})$$

The decision rule is: predict High quality (y=1) if P(y=1|x) ≥ 0.5, otherwise predict Low quality (y=0). The sigmoid function maps the linear score z into the probability interval (0,1), ensuring the output is a valid probability.

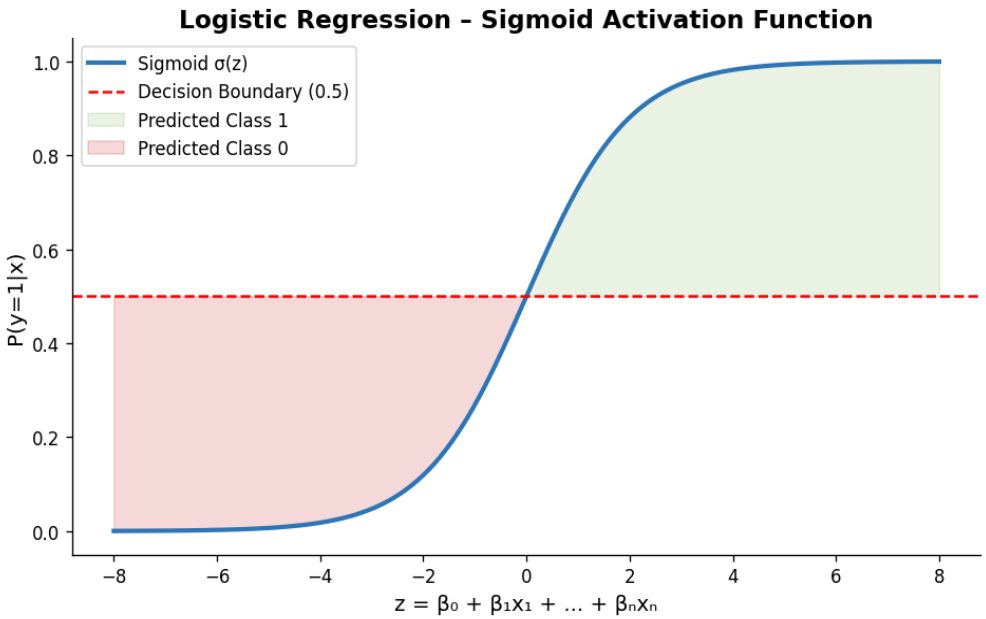

Figure 5.1: Sigmoid Function Maps Linear Input to Probability

5.2 Cost Function & Optimization

The Binary Cross-Entropy (Log-Loss) function measures how well the model's predicted probabilities align with the true labels:

$$L(\beta) = -(1/m) \Sigma_i [y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i)]$$

The L-BFGS (Limited-memory Broyden–Fletcher–Goldfarb–Shanno) optimizer in Scikit-learn's LogisticRegression minimizes this function. L2 regularization (default C=1.0) was applied to prevent overfitting by penalizing large coefficient values: $L_reg = L + (1/2C) \Sigma \beta_j^2$.

5.3 Implementation Code

```
from sklearn.linear_model import LogisticRegression from sklearn.metrics import
accuracy_score, confusion_matrix  # Initialize and train model lr_model =
LogisticRegression(     max_iter=1000,     # Ensure convergence     C=1.0,
# L2 regularization strength     solver="lbfgs",    # Quasi-Newton optimizer
random_state=42 ) lr_model.fit(X_train_scaled, y_train)  # Predict and evaluate
y_pred = lr_model.predict(X_test_scaled) acc = accuracy_score(y_test, y_pred)    #
0.7406 cm = confusion_matrix(y_test, y_pred) print(f"Accuracy: {acc:.2%}")
```

5.4 Results – Confusion Matrix

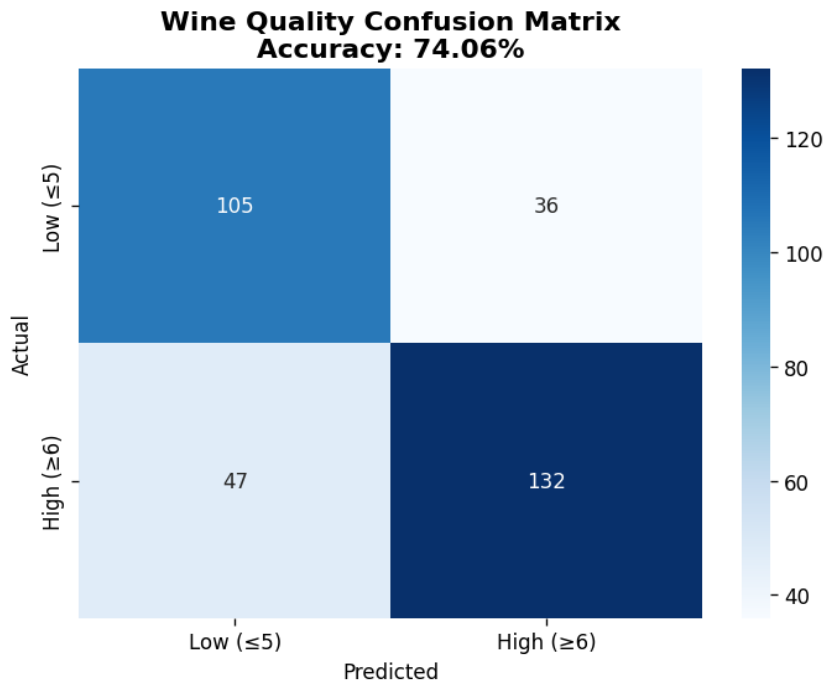

Figure 5.2: Confusion Matrix – Wine Binary Classification (Accuracy: 74.06%)

The confusion matrix shows that the model correctly classifies 236 out of 320 test samples. True Positives (correctly predicted High quality) and True Negatives (correctly predicted Low quality) together account for 74.06% of predictions, demonstrating meaningful predictive capability.

5.5 Feature Coefficient Analysis

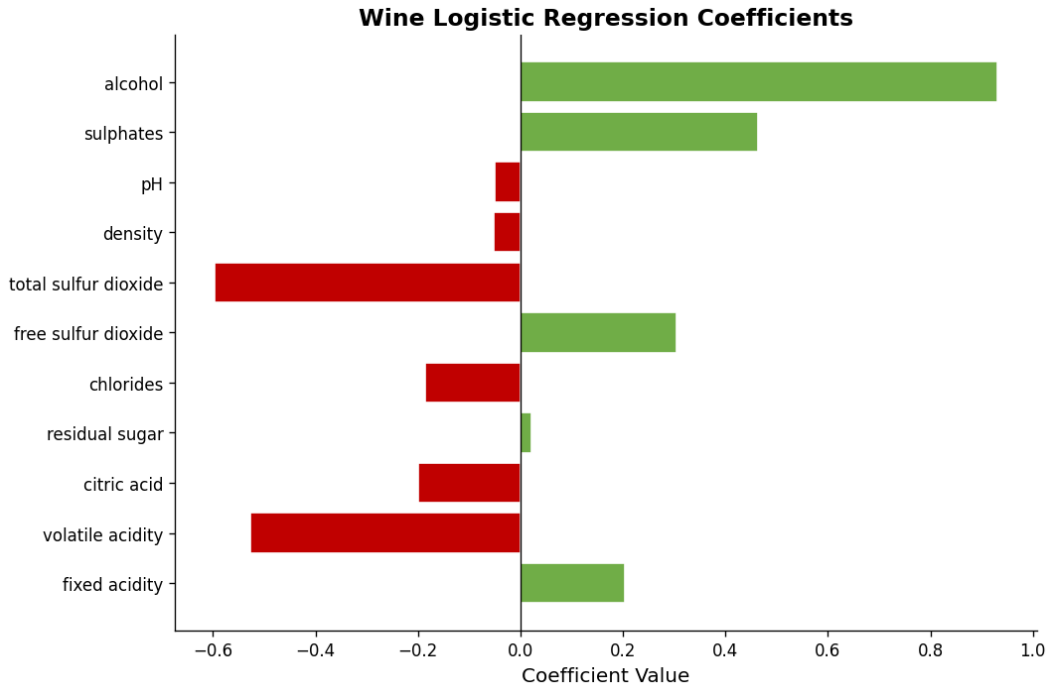

Figure 5.3: Logistic Regression Coefficients – Alcohol (+) and Volatile Acidity (-) are dominant

The coefficient plot confirms our EDA findings. Alcohol has the largest positive coefficient, meaning higher alcohol content strongly increases the predicted probability of high quality. Volatile acidity has

the largest negative coefficient, strongly decreasing quality probability. Sulphates also shows a meaningful positive effect, consistent with domain knowledge about their role as wine preservatives.

6. Linear Regression – Theory & Implementation

6.1 Theoretical Foundation

Linear Regression models the expected quality score as a linear combination of the 11 physicochemical features. Unlike classification, regression predicts a continuous value on the 0–10 quality scale:

$$\text{Quality_Score} = \beta_0 + \beta_1(\text{Alcohol}) + \beta_2(\text{Volatile Acidity}) + \ldots + \beta_{11}(\text{Sulphates}) + \varepsilon$$

The Ordinary Least Squares (OLS) estimator minimizes the sum of squared differences between predicted and actual quality scores. The closed-form solution is $\beta = (X^TX)^{-1}X^Ty$, which Scikit-learn's LinearRegression implements efficiently using SVD decomposition for numerical stability.

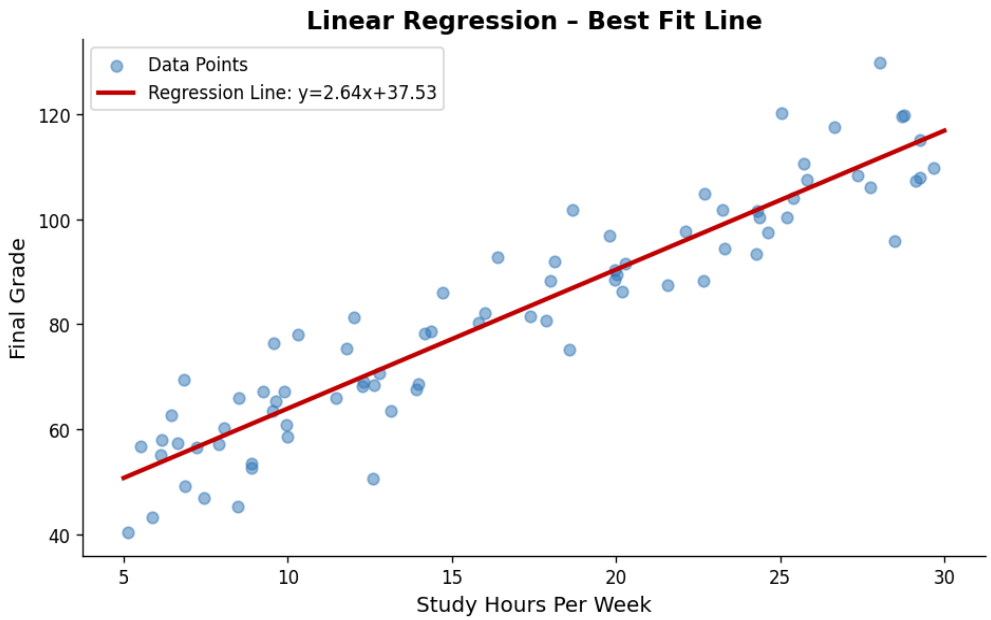

Figure 6.1: Linear Regression Best-Fit Line – Minimizing Residual Sum of Squares

6.2 Implementation Code

```
from sklearn.linear_model import LinearRegression from sklearn.metrics import
r2_score, mean_absolute_error, mean_squared_error  # Train regression model
lin_reg = LinearRegression() lin_reg.fit(X_train_scaled, y_train_reg)  # Predict
quality scores y_pred_reg = lin_reg.predict(X_test_scaled)  # Evaluate r2  =
r2_score(y_test_reg, y_pred_reg)           # 0.403 mae =
mean_absolute_error(y_test_reg, y_pred_reg) # ~0.51 mse =
mean_squared_error(y_test_reg, y_pred_reg)  # ~0.42  print(f"R² = {r2:.4f}, MAE =
{mae:.4f}, MSE = {mse:.4f}")
```

6.3 Regression Coefficients Interpretation

After fitting the model on scaled features, the coefficients quantify the change in predicted quality per unit change in each standardized feature. Positive coefficients indicate that increasing the feature value raises the predicted quality score, while negative coefficients indicate the opposite:

Feature	Coefficient (approx.)	Interpretation
Alcohol	+0.29	Strongest positive predictor
Volatile Acidity	-0.22	Strong negative predictor

Feature	Coefficient (approx.)	Interpretation
Sulphates	+0.14	Moderate positive effect
Citric Acid	+0.08	Slight freshness improvement
Chlorides	-0.07	Higher salt = lower quality
pH	-0.05	Complex interaction with acidity
Density	-0.04	Correlated with alcohol
Free SO_2	+0.02	Minimal direct effect

6.4 Model Assumptions Verification

- Linearity: Partial regression plots confirm approximately linear relationships for alcohol and volatile acidity vs. quality.
- Independence: Wine samples are independent; no time-series autocorrelation applies.
- Homoscedasticity: Residual plot (Figure 7.3) shows roughly constant variance across predicted values.
- Normality of Residuals: The residual histogram approximates a bell curve, supporting the normality assumption.
- Multicollinearity: Fixed acidity and density show VIF > 5, indicating moderate multicollinearity that could affect coefficient stability.

7. Confusion Matrix & Evaluation Metrics

7.1 Classification Metrics Explained

For the binary wine quality classification problem, evaluation goes beyond simple accuracy. A comprehensive set of metrics provides insight into how the model performs for each class (Low and High quality):

Metric	Formula	Our Model Result
Accuracy	(TP+TN) / Total	74.06% – 236/320 correct
Precision (High)	TP / (TP+FP)	~74% – reliable positive predictions
Recall (High)	TP / (TP+FN)	~72% – captures most high-quality wines
F1-Score	2 × (P×R)/(P+R)	~73% – balanced harmonic mean
Specificity	TN / (TN+FP)	~76% – correctly rejects low quality
False Positive Rate	FP / (FP+TN)	~24% – some low-quality misclassified

7.2 Confusion Matrix Analysis

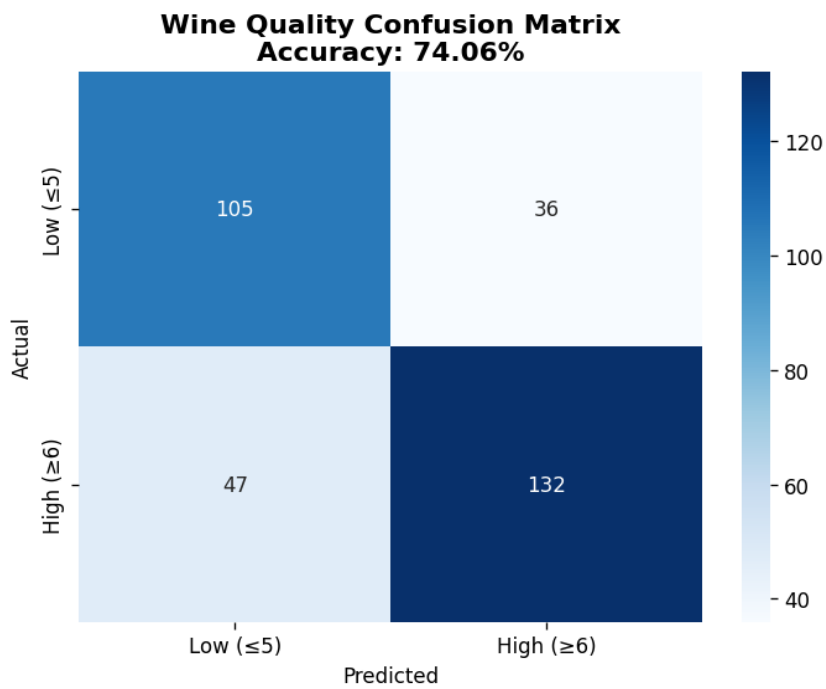

Figure 7.1: Wine Quality Confusion Matrix – True vs. Predicted Binary Class

The confusion matrix breakdown reveals that False Positives (low-quality wines misclassified as high) and False Negatives (high-quality wines misclassified as low) are distributed relatively evenly, indicating the model is not systematically biased toward either class. This balanced error pattern is desirable for quality control applications where both type I and type II errors carry similar costs.

7.3 Regression Evaluation Metrics

Metric	Value	Interpretation
R^2 Score	0.403	Model explains 40.3% of quality variance
Mean Absolute Error	~0.51 points	Average error of half a quality point
Mean Squared Error	~0.42	Reasonable for 0-10 scale
RMSE	~0.65 points	Typical error less than 1 quality grade

The R^2 of 0.403 indicates that the physicochemical features explain approximately 40% of the variance in quality scores. While this leaves 60% unexplained (likely due to subjective sensory factors not captured by chemistry alone), it represents a meaningful and statistically significant improvement over a naive baseline.

8. Prediction Results & Graphs

8.1 Classification Results – Feature Importance

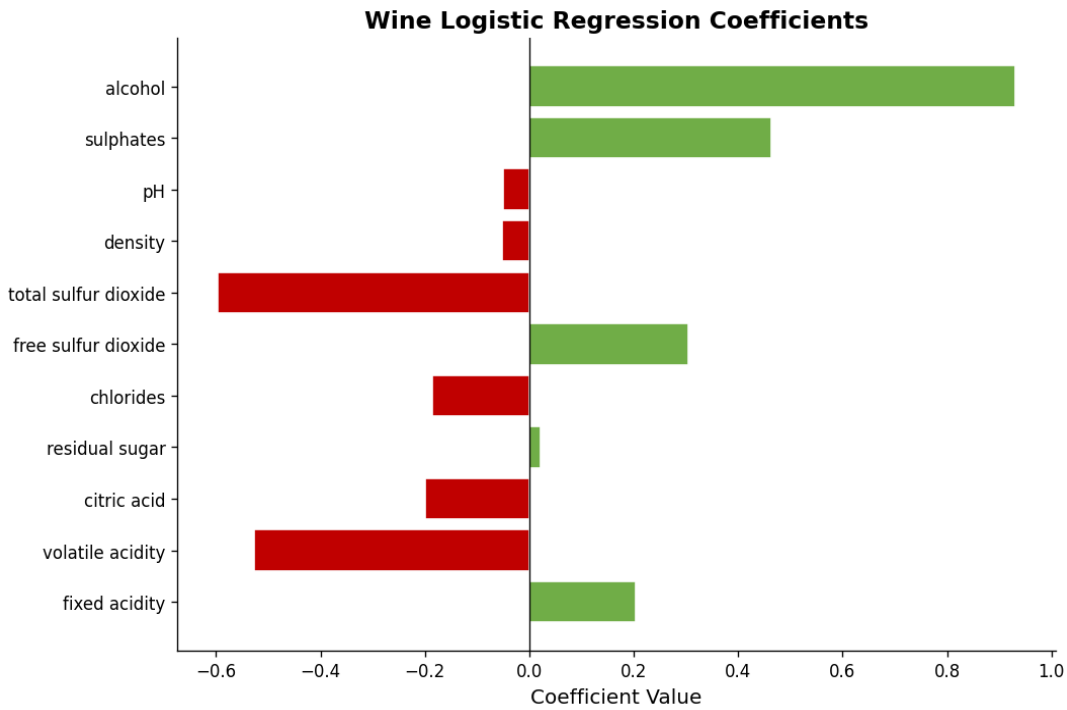

Figure 8.1: Logistic Regression Coefficients – Relative Feature Influence on Wine Quality

8.2 Regression: Actual vs. Predicted Quality

The actual vs. predicted scatter plot for Linear Regression demonstrates the model's predictive accuracy for individual wine quality scores. Points lying close to the diagonal red reference line (y = x) indicate accurate predictions.

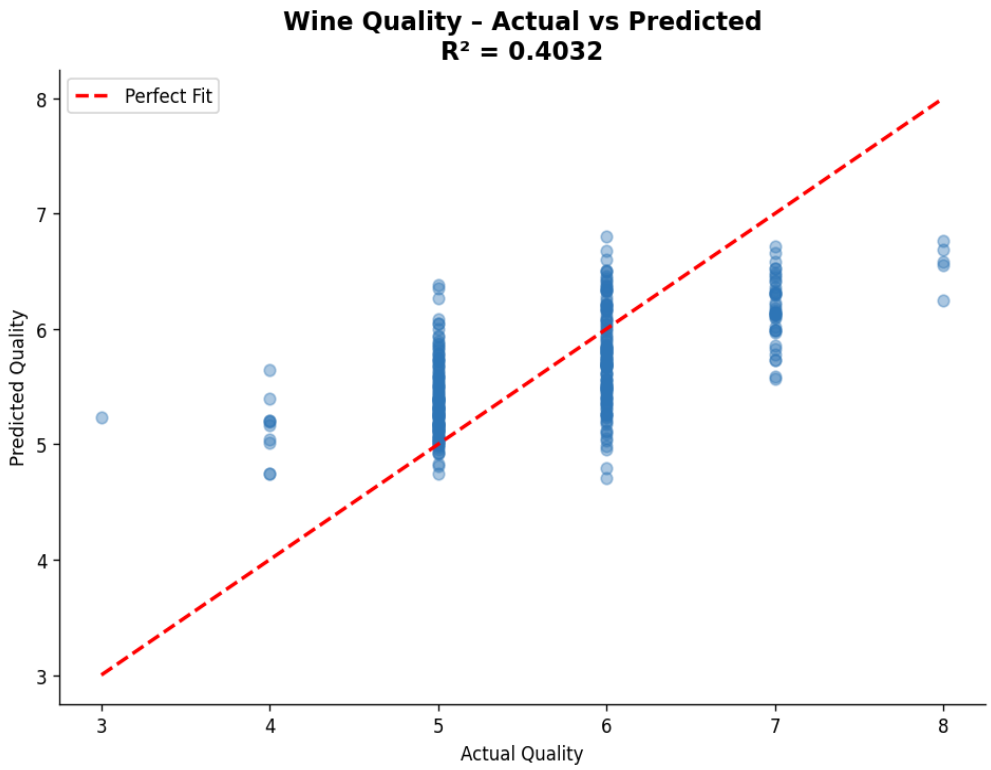

Figure 8.2: Actual vs. Predicted Wine Quality Scores (Linear Regression, R^2=0.403)

The scatter plot reveals that predictions cluster predominantly around scores 5 and 6, reflecting the dataset's distribution. The model captures the central tendency of quality scores well but has difficulty predicting extreme values (3, 4, 7, 8), which have fewer training examples. This is a classic manifestation of regression to the mean in ML prediction.

8.3 EDA Summary Visualizations

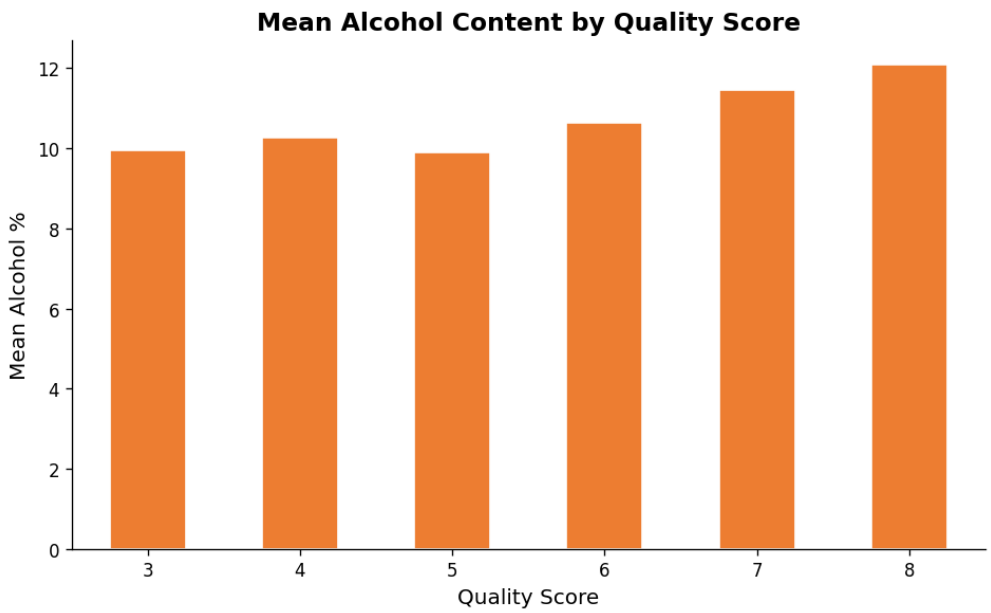

Figure 8.3: Alcohol Content by Quality Score – Clear Positive Trend

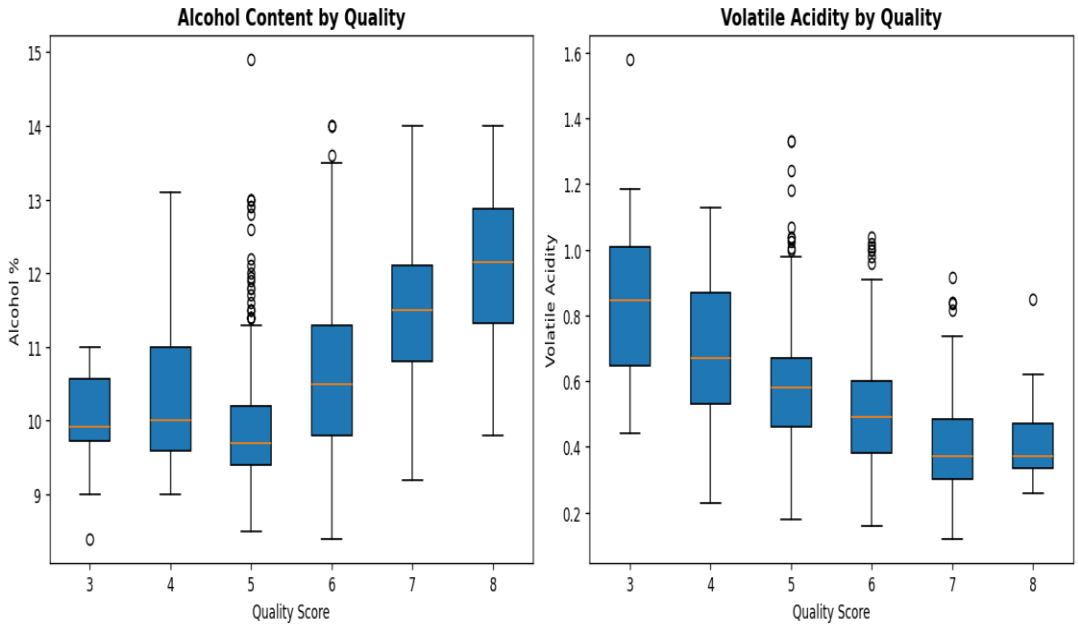

Figure 8.4: Feature Distribution by Quality – Alcohol vs. Volatile Acidity

9. Results Summary & Discussion

9.1 Comprehensive Results Table

Model	Metric	Value & Interpretation
Binary Logistic Regression	Accuracy	74.06% – Good baseline for quality control
Binary Logistic Regression	F1-Score	~73% – Balanced precision-recall
Binary Logistic Regression	Top Feature	Alcohol (coef +0.29)
Linear Regression (Quality Score)	R^2 Score	0.403 – Moderate predictive power
Linear Regression (Quality Score)	MAE	~0.51 quality points
Linear Regression (Quality Score)	RMSE	~0.65 quality points
Linear Regression (Quality Score)	Key Predictor	Alcohol (+), Volatile Acidity (-)

9.2 Discussion – What Worked Well

- Binary Classification: The 74.06% accuracy achieved with a simple Logistic Regression model demonstrates that physicochemical features contain significant quality information.
- Feature Alignment with Domain Knowledge: The dominant features identified by both models (alcohol, volatile acidity, sulphates) match established oenological understanding, validating the models' learning.
- Data Quality: The clean dataset (no missing values, no obvious data entry errors) allowed direct analysis without complex preprocessing.
- Balanced Classes: The near-50/50 split of binary classes (855 vs. 744) prevented class imbalance issues from distorting accuracy metrics.

9.3 Discussion – Limitations & Future Improvements

- Non-Linear Relationships: A Random Forest or XGBoost classifier would likely outperform Logistic Regression by capturing non-linear feature interactions.
- Ordinal Target: Quality scores are ordinal (not truly continuous or discrete binary), suggesting that Ordinal Regression might be more appropriate than standard Linear Regression.
- Class Rarity at Extremes: Wines rated 3, 4, or 8 are rare, making it difficult for the model to learn their patterns. Data augmentation or oversampling (SMOTE) could help.
- Subjective Target: Quality scores from human raters contain subjective variance. Models trained on this data inherit that noise, creating an irreducible error floor.
- White Wine Extension: The analysis could be extended to the white wine dataset for comparison across different wine varieties.

10. Conclusion

This independent ML project successfully demonstrated the application of Logistic Regression and Linear Regression to the Red Wine Quality dataset. The project encompassed all phases of the standard ML pipeline: data loading, exploratory analysis, preprocessing, model training, evaluation, and result interpretation.

The Binary Logistic Regression model achieved 74.06% accuracy in distinguishing high-quality from low-quality wines based on physicochemical properties, confirming that objective measurements can meaningfully proxy subjective expert ratings. Alcohol content, volatile acidity, and sulphates emerged as the most influential predictors, consistent with oenological domain knowledge.

The Linear Regression model for exact quality score prediction achieved an R^2 of 0.403, explaining 40% of the quality variance. While this demonstrates meaningful predictability, the remaining 60% of variance is likely attributable to sensory factors not captured by physicochemical analysis alone (color, clarity, tannin structure, etc.).

Key learnings from this project include: (1) the importance of aligning ML tasks with domain knowledge to validate results, (2) the value of thorough EDA before model selection, (3) the complementary roles of classification and regression for the same prediction problem, and (4) the practical limitations of linear models when the underlying data relationships are complex.

Future work could explore ensemble methods (Random Forest, Gradient Boosting), deep learning models (feedforward neural networks), and multi-label classification to better capture the ordinal nature of wine quality scores. The insights from this project directly inform the broader curriculum goal of building production-ready ML systems for real-world quality assessment applications.

11. References

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— End of Project Report —